WEEK 5 · LECTURE L2 OF 3 · UNIT II

Learning Management Systems

From course catalogue to skills platform, and what makes learning change behavior.

Part 1	How adults learn; why micro, spaced and mobile
Part 2	LMS, LXP and skills platforms; the Infosys Lex case
Part 3	A learning path for a store associate; Kirkpatrick and transfer
Read	Case: Infosys Lex; NASSCOM FutureSkills, selected sections

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PART C · 1:15 TO 2:00

Learning systems should change behavior, not count completions

How do adults learn, and what follows for design?

Why micro and spaced learning?

What did GreenLeaf’s data say about training?

PART C · THEORY 8

How adults learn: two theories set the design brief

Knowles: andragogy (1980)

Adults need to know why.

They direct their own learning.

Experience is the main resource.

They learn around real problems.

Kolb: experiential cycle (1984)

1 Concrete experience

2 Reflective observation

3 Abstract conceptualization

4 Active experimentation

Caution: matching teaching to “learning styles” has no good evidence (Pashler et al., 2008).

PART C · THEORY 8b

Self determination theory: why points and badges backfire

Three basic needs

Deci and Ryan (1985); Ryan and Deci (2000).

Autonomy, **competence** and **relatedness** sustain intrinsic motivation.

Overjustification

Deci, Koestner and Ryan (1999), meta analysis: expected tangible rewards undermined intrinsic motivation for interesting tasks.

Informational feedback did not.

Design consequences

Choice of path (autonomy).

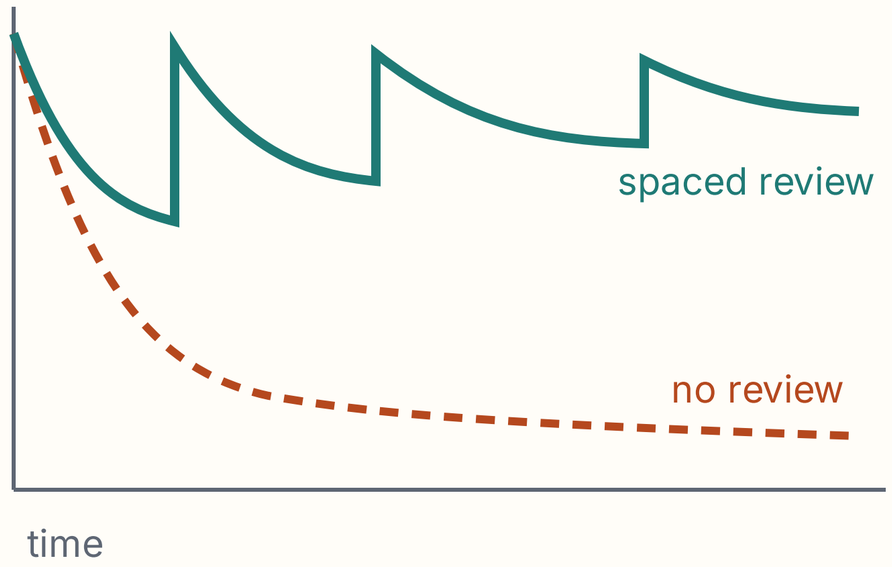
Visible skill growth (competence).

Learning with peers (relatedness).

This is the theory behind the Round 3 caution: extrinsic rewards create activity without capability.

PART C · THEORY 9

Why micro and spaced: the forgetting curve



Three findings

Ebbinghaus (1885): memory fades fast without review.

Spacing effect: spread practice beats one long session (Cepeda et al., 2006).

Mayer: short segments, clear signals, no clutter.

The curve is illustrative, not data. Mayer, R. E. (2009). Multimedia Learning, 2nd ed. Cambridge University Press.

PART C · THEORY 9b

Desirable difficulties: what makes learning stick

Principle	Finding	Mobile design at GreenLeaf
Spacing	Spread practice beats massed (Cepeda et al., 2006)	Short refreshers on days 2, 7 and 21
Retrieval practice	Testing beats rereading (Roediger and Karpicke, 2006)	Quiz before the content, not after
Interleaving	Mixed problems beat blocked ones (Rohrer and Taylor, 2007)	Mix leave, payslip and shift tasks
Segmenting	Learner paced short parts help novices (Mayer, 2009)	One task per two minute clip

Bjork, R. A. (1994). Memory and metamemory considerations in the training of human beings.

PART C · PLATFORMS

LMS, LXP, skills platform: three different logics

	LMS	LXP	Skills platform
Organizing unit	Course	Content experience	Skill
Who chooses	HR assigns	Learner browses	Role gap drives
Main metric	Completion	Engagement	Proficiency gain
Best GreenLeaf use	Compliance	Corporate staff	Frontline reskilling

Case: Infosys Lex. As you read it, ask which organizing unit Lex chose, and what that choice made measurable.

L2 · CASE

Infosys Lex: digital learning at scale

What it is

Infosys's in-house learning platform, launched for its own employees in April 2018. Cloud first and mobile first.

114,000+

Scale

Used by more than 114,000 Infosys employees within months of launch.

How it works

Machine-learning course suggestions, learning playlists, and content that employees create and share.

Then sold outside

In September 2018 Infosys began offering the same platform to client firms as Infosys Wingspan.

The question for GreenLeaf

Infosys staff are graduates on laptops who read English. What would have to change for Lakshmi Naidu?

SOURCE Infosys press release, 20 September 2018. Transfer the design principle, not the product.

L2 · DESIGN EXERCISE

A first-month learning path for a GreenLeaf store associate

Week	Course in GreenLearn	Design choice
1	Using the GreenLeaf ESS App	Five-minute videos in the associate's language, on the phone
2	POSH Compliance	Short scenarios, not a long text module
3	Customer Service Excellence	Practiced on the floor with a buddy (blended)
4	Inventory Systems	Hands-on with the store's scanner, then a short check
Ongoing	Data Privacy and DPDP Basics	Spaced refreshers, not one sitting

THE EVIDENCE

Lakshmi's one course, POSH, is In-progress with a score of 32%. Smartphone-only staff complete 61% of courses; laptop users 90%. Source: LMS_Learning joined to Employee_Master.

PART C · THEORY 10

Kirkpatrick’s four levels, mapped to GreenLeaf’s tabs

4	**Results**	Helpdesk load, attrition, store KPIs	**PeopleAnalytics_Fact**
3	**Behavior**	ESS logins and self service use	**Adoption_Telemetry**
2	**Learning**	Assessment score, completion	**LMS_Learning**
1	**Reaction**	Did learners value it?	**Not collected**

Kirkpatrick (1959); Kirkpatrick and Kirkpatrick (2016), Kirkpatrick’s Four Levels of Training Evaluation, ATD Press.

PART C · CRITIQUE

The four levels are not a causal chain

Alliger et al. (1997)

Meta analysis: reactions correlated only weakly with learning and with on the job behavior.

Judgments of usefulness did better than enjoyment.

Holton (1996)

The model is a taxonomy, not a theory: it names outcomes without explaining why one leads to another.

Context and motivation are left out.

So a high Level 2 score licenses no claim about Level 3. You must measure behavior directly, which GreenLeaf’s data allows.

PART C · THEORY 10b

Transfer of training: the missing causal model

Inputs

Trainee: ability, motivation
Design: practice, relevance
Work environment: support, chance to use

→

Learning

Learned and retained

→

Transfer

Used on the job and kept up

Blume et al. (2010) meta analysis: work environment support and motivation relate to transfer. A module cannot supply the store manager who lets you use what you learned.

Baldwin, T. T. and Ford, J. K. (1988). Transfer of training: A review and directions for future research. Personnel Psychology, 41(1).

Going deeper: the theory underneath learning systems

The next slides set out the 70-20-10 idea and its weak evidence, and Phillips on return on investment: what each claims, how it works, where it breaks, and what it means at GreenLeaf.

THEORY IN DEPTH · LOMBARDO AND EICHINGER (1996)

70-20-10: popular, but weakly evidenced

THE CORE IDEA

The claim: about 70% of learning comes from work, 20% from others, 10% from formal courses.

HOW IT WORKS

It pushes firms to design learning into the job, with coaching and stretch tasks.

LIMITS AND CRITIQUE

The ratio comes from executive self-reports, not measurement; treat it as a reminder, not a fact.

AT GREENLEAF

For a store associate, the shift and the buddy are most of the curriculum. The LMS supports them.

THEORY IN DEPTH · PHILLIPS (1997)

Add a fifth level: return on investment

THE CORE IDEA

Phillips extended Kirkpatrick with a fifth level that compares money benefits with costs.

HOW IT WORKS

Isolate the training effect, convert it to money, and divide by the full cost.

LIMITS AND CRITIQUE

Isolating the effect is hard; ROI figures are often more precise than the evidence allows.

AT GREENLEAF

Before claiming ROI for GreenLearn, show that behavior changed. Completion alone is level one.